

TUTORLINKAI: A Verified Local Tutor Discovery and Booking Platform

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Abstract- In the fast-paced world, education plays a vital role in shaping the future of the younger generation. Many students find it difficult to keep up with their classes and compete academically with their peers. This creates a need for additional academic support for the students, leading to parents seeking tutors who can help their children understand the concepts where they require more assistance.

Parents often choose tutors who are close to their homes or tutors who have been suggested by relatives or neighbours; they miss out on checking if the tutor is the right fit for their child.

It is given that the parents want what is best for their child and help them academically, but they do not always perform background checks and assess whether this tutor is appropriate for the needs of their child and if they are properly qualified.

This often leads to the child not being able to understand the concepts despite the parents providing them with some help, but just not the right resources.

Another major problem that parents and students face is the lack of verification for the tutors. We must understand who is helping our children and assess if that person is legitimate and trustworthy. It is also important that we perform a credibility check rather than blindly believing the neighbors' or relatives' suggestions.

This is where TutorLinkAI comes in; the main goal of this application is to provide parents with the right information

to help provide the children with the right resources that help them academically without adding any strain.

TutorLinkAI focuses on connecting parents with the right tutors by assessing their qualifications and seeing if they can help in subjects where the child needs help.

Using TutorLinkAI, parents can check if the tutor is actually credible and has the appropriate degrees and credentials to help their children; this creates a sense of relief for both the children and the parents.

Keywords: TutorLinkAI, Online Tutoring, Tutor Verification, Credential Verification, Optical Character Recognition (OCR), Artificial Intelligence, Tutor Recommendation, Educational Technology, Academic Support, Trust and Safety

I. INTRODUCTION

Learning has increasingly moved beyond the confines of a traditional classroom. Given the academic background of students and their level of comprehension, classroom teaching may not always be enough and as such tuition services have long provided additional support to students who require more help in certain areas of learning than what is provided in the classroom. However, the efficiency of

such assistance depends on locating a tutor who is qualified, experienced, and able to provide guidance to the student according to their specific needs.

Traditionally, students and parents have relied on a number of private networks, including family, friends, neighbours in order to locate tutors. While such methods can be effective in shortlisting potential tutors, the information provided is usually limited. Factors to consider when selecting a tutor include qualifications, the subject matter or area of expertise that the student needs help with, experience, location, availability, and fees amongst other considerations. Gathering and comparing this information can be a tedious process which many parents and students find inconvenient.

Online services have been developed to enable users to first research and then locate a suitable service based on certain criteria. However, there are various limitations associated with existing tutor discovery methods and services. Firstly, many existing platforms do not allow parents or students to easily locate tutors using certain criteria, as well as comparing tutors based on certain requirements. Such an option can prove especially useful to parents and students looking for specific tutors. In many cases, finding a tutor relies heavily on word of mouth, which limits students and parents to tutors within their personal networks and provides limited opportunities to match tutors effectively with the specific needs and requirements of the student.

Even further into the tutor discovery process, parents and students must engage in separate communication channels when contacting a tutor. As such, parents and students must individually contact a tutor in order to confirm their qualifications, availability, as well as schedule and reschedule sessions. This can be inconvenient and in many cases redundant. A proper platform can enable both students and parents to communicate with tutors and manage scheduling and other activities related to the tutoring sessions.

An equally important limitation in the tutor discovery process is the reliability of information provided to parents and students. It is vital that both students and parents consider a number of factors when selecting a tutor, including educational background, experience, and the area of expertise. Making such information easily accessible to parents and students can be extremely valuable in ensuring that both parties have realistic expectations of the tutor's qualifications and ability.

The proposed Local Tuition Discovery Platform, TutorLinkAI, is aimed at addressing the various limitations of the traditional methods of tutor discovery. By aggregating tutor related information and offering a wide range of features for both parents and tutors, TutorLinkAI facilitates

the tutor discovery process based on a number of criteria including subject, location, qualifications, availability, experience and affordability. By providing tutors with the ability to post their information and experience for students and parents to view, the proposed solution addresses the limitations of relying on word of mouth by offering both parents and students a wider pool of tutors to consider. Additionally, the proposed platform enables students and parents to sort through the available tutors according to their specific needs, thus eliminating the need to manually sift through hundreds of tutors or advertisements.

The proposed Local Tuition Discovery Platform, TutorLinkAI, supports the United Nations Sustainable Development Goal (SDG) 4: Quality Education by enabling students to receive more personalized supplementary education. By enabling proper matching of students and tutors based on student requirements as well as tutor qualifications, which focus on enhancing personalized learning and promoting access to quality education for all.

Apart from supporting universal access to quality education, TutorLinkAI aligns with SDG 10: Reduced Inequalities, which emphasizes the need for empowering economically disadvantaged individuals. By providing affordable means of accessing supplementary education and offering students and parents the option of sorting tutors based on their budget, TutorLinkAI can significantly help reduce the disparities associated with access to additional academic services.

As such, the proposed Local Tuition Discovery Platform, TutorLinkAI, is aimed at enabling personalized education by addressing the inefficiencies associated with accessing information regarding local tuition services. By incorporating local tuition services, offering additional information on tutors' qualifications and experience, as well as providing various tools for filtering and matching tutors with students based on their specific needs and budgets, TutorLinkAI facilitates more convenience and transparency in locating potential tutors.

II. LITERATURE SURVEY

The literature survey provides a theoretical framework and identifies prior research relevant to the design and development of intelligent online tutor discovery and matching services. Existing solutions primarily aim to bridge the gap between students and suitable tutors through subject-based searches, location-based discovery and booking facilities. The early approaches depend predominantly on the profile information provided by the user and predefined search categories, providing limited support for personalised and student-centric tutor recommendations [6]-[8].

Several online tutor discovery systems have investigated structured student-tutor management and filtering mechanisms. These applications have incorporated criteria such as education level, subject expertise and geographical location for tutor filtering. A comparative analysis of existing applications student and tutor registration and tutor filtering are common functionalities, while features such as tutor-parent interaction and communication, tutor ranking provide additional support. These approaches demonstrate the feasibility of recommendation-based tutor discovery while also indicating the importance of incorporating multiple learner and tutor characteristics [2].

Location-based approaches have improved tutor discovery by incorporating geographical context such as latitude and longitude information to map tutor and student locations and filter tutors based on proximity. This is particularly practical in cases where travelling distance is a priority. Beyond online tutor-finding applications, online location-based service recommendation systems provide advanced mechanisms to combine proximity with other attributes. These systems employ mechanisms such as the Haversine formula to calculate geographical distance and use K-Nearest Neighbours (KNN) with scoring mechanisms based on factors such as proximity, price and ratings to rank service providers. These systems support service provider availability, booking and reviews. Such approaches demonstrate that geographical proximity can be incorporated with multiple other factors to determine the most suitable service provider. However, proximity alone is not sufficient to determine academic suitability and can be combined with other features, such as subject expertise, tutor reviews, and teaching approach, to identify a suitable tutor for a student's particular learning needs.

The limitations of static filtering have led to the development of personalized recommendation systems that incorporate user preferences along with behavioral information and service provider characteristics to generate relevant recommendations[6], [14]. Machine learning based recommendation systems use historical data such as user-provider relationships and combine it with contextual re-ranking and factors such as urgency, provider availability, ratings and user interactions. This demonstrates how such systems can move beyond simple filtering by learning user-provider relationships. For the tutor recommendation system the same principle can be used by considering students and parents as users and tutors as education providers. Additional features can be incorporated, including subject and topic expertise, academic level, student skill requirements, teaching preferences, and learning objectives.

An important extension to conventional tutor recommendation systems is the incorporation of student skill and knowledge assessment. A student may identify a subject as a requirement without being able to accurately identify the specific concepts in which additional assistance is required. Knowledge-tracing techniques provide a mechanism for addressing this issue by estimating a learner's evolving knowledge state from historical learning interactions [1]. One such approach is a Transformer-based Deep Knowledge Tracing framework for formative assessment of student skills [9]. The framework analyses sequences of student interactions to estimate knowledge states and identify knowledge gaps. The resulting learner representation can provide valuable information for tutor recommendation because it can distinguish between a student needing assistance across the entire subject and a student who specifically requires help in specific topics. Although knowledge-tracing research primarily focuses on student assessment, its resulting learner representation can be used as an input to identify suitable tutors.

Technical approaches such as semantic representation of textual information provide another means of improving recommendation systems beyond keyword matching. Sentence embedding models represent textual information in vector spaces where semantic relationships are measured computationally. Ni et al. proposed Sentence- T5 for scalable sentence encoding and evaluated its effectiveness for semantic textual similarity tasks [10]. This approach provides a basis for comparing descriptive student requirements with tutor expertise, particularly when different terms are used to express related concepts.

Automated document analysis can further support credential verification by identifying potentially manipulated documents. Al-Ameri et al. investigated an unsupervised method for detecting document forgery using document analysis techniques [3]. Such approaches demonstrate that document-level analysis can complement text extraction and profile consistency checks. In the current TutorLinkAI implementation, OCR is primarily used to extract information from the submitted certificates, while advanced document forgery detection is a potential future extension.

The literature therefore indicates that tutor discovery can be viewed as a multi-criteria recommendation problem. Educational compatibility can be determined from factors such as student skill requirements, subject or topic preferences and tutor skills and expertise, while contextual feasibility can be incorporated through factors such as pricing, availability, distance, ratings and tutor profile verification. Existing tutor recommendation systems demonstrate the importance of profile-based matching, location-based systems demonstrate the usefulness of

geographical and service constraints, machine-learning-based recommendation systems highlight the importance of behavioural and contextual personalization, and knowledge-tracing research highlights the importance of student assessment for identifying knowledge levels and mapping student needs. However, these capabilities are generally addressed as separate problems rather than being integrated into a unified tutor recommendation process.

III. EXISTING SYSTEM AND PROBLEM IDENTIFICATION

Existing solutions are mostly old-fashioned coaching institutes where their main goal is to increase student enrollment every year. This causes a financial drain on families and prevents students from finding the right help. These coaching institutes often take classes for 20-30 students at once, leading to a lack of personalisation and failing to cater to the individual needs of each student.

Most centers also assign the students homework assignments, adding to the physical and mental exhaustion of the students, which in turn again leads to them not performing as expected academically.

Although one can argue the demerits of the existing traditional methods, it is also important that we recognise the few centers that actually upskill the students and help them improve their academic capabilities through their extensive coaching.

Some existing online tutor finder platforms map the students to tutors who reside in a different pincode; this adds to the physical exhaustion of the students.

The platforms often advertise their fancy courses, which usually end up as clickbait for increased enrollments; additionally, students having to choose from a wide variety of courses and faculties can become hectic.

Students usually have limited information about the availability of tutors in their localities, tutors who can help them in the subjects they are struggling with, and help them find tutors who charge fees that are economical for them.

Major Drawbacks of Existing Solutions

There are several ways for students to find tutors, but most of the existing methods still have some drawbacks.

1. Commercial Coaching Institutes and Manual Tuition Centers

In most cases, students find local tuition centers or coaching classes through advertisements, recommendations, or by asking around. Often the classes can end up being expensive, or they can be taught in groups, hence leaving the tutor to not be able to cater to individual student needs.

Finding the right tutor can also take time because parents often have to make calls or visit different centres.

2. Web and Directory-Based Tutor Portals

Examples: A+ Home Tuition, bdtutors.com and similar websites.

Online websites let students search for tutors using basic details like subject and location, and do not focus on important factors like student needs, affordability, and class scheduling. These platforms do not have direct communication between the tutors and the parents, and hence arranging classes may take extra time. The matching is also mostly based on profile details rather than what the student actually needs help with.

3. Multi-Criteria and Location-Based Platforms

Examples: GRAB Tutor Management System (GTMS), Tutors Map and Tutor Finder (LBS).

These platforms allow students to search for tutors using filters based on subject, gender, and expertise. They mainly use the information given in the tutor's profile. These platforms do not consider factors such as students' weak areas, teaching quality, availability, and fees.

Overall, most existing solutions focus on finding tutors using basic information or past activity. They do not give enough importance to the student's actual learning difficulties, current performance, tutor availability, and cost.

IV. PROPOSED METHODOLOGY

The application TutorLinkAI is a secure and intelligent platform that is designed to help parents and students in identifying the suitable and credible tutors. The application addresses two of the major limitations in the existing tutor selection system, i.e., the difficulty of assessing the tutor suitability and the lack of reliable tutor verification in the existing systems.

The proposed system consists of a tutor onboarding module, credential verification module, tutor profile management module, and tutor discovery and matching module. During onboarding, tutors register on the platform by providing their personal, academic, and professional information. An authentication mechanism is used to verify the tutor's account before the profile is created.

A key component of TutorLinkAI is the tutor credential verification process. Tutors are required to upload relevant qualification documents as part of the onboarding process. The uploaded documents are processed using Optical Character Recognition (OCR) to extract relevant textual

information. The extracted information is then analyzed and compared with the information provided by the tutor during registration. Similarity-based validation is used to identify inconsistencies between the submitted credentials and the registered profile information. Based on the verification results, the system identifies whether the submitted information is consistent with the registered tutor profile and assigns an appropriate verification status.

The backend of the application is implemented using Python and FastAPI, this provides RESTful APIs for authentication, onboarding, profile management, document processing and verification. PostgreSQL is used for persistent data storage, with SQLAlchemy providing database interaction and Alembic supporting database schema migration. The frontend provides interfaces for tutor registration, profile creation, credential submission, and tutor discovery.

In the application, once the tutors are verified, the students and parents can view their profiles and identify tutors based on relevant academic requirements. The application combines identity onboarding, credential verification and tutor suitability assessment within one platform, hence providing a structured alternative to relying solely on recommendations from relatives, neighbors or nearby tutors.

V. SYSTEM DESIGN AND IMPLEMENTATION

The proposed TutorLinkAI system is designed as a modular intelligent platform for tutor discovery and personalized tutor recommendation. The system integrates student and tutor profiling, tutor verification, student requirement assessment, AI-based matching, and booking within a unified platform. The architecture separates the user interface, application logic, data management, and AI-based processing components to support modular development and future extension.

A. System Architecture

TutorLinkAI follows a modular client-server architecture consisting of a React-based frontend, FastAPI backend, PostgreSQL database and dedicated AI services. The frontend provides interfaces for student and tutor registration, profile management, assessment, tutor discovery, recommendation, and booking. The backend manages authentication, application logic, database operations and communication between the frontend and AI services. AI services handle computationally intensive operations such as semantic matching, face verification and certificate OCR.

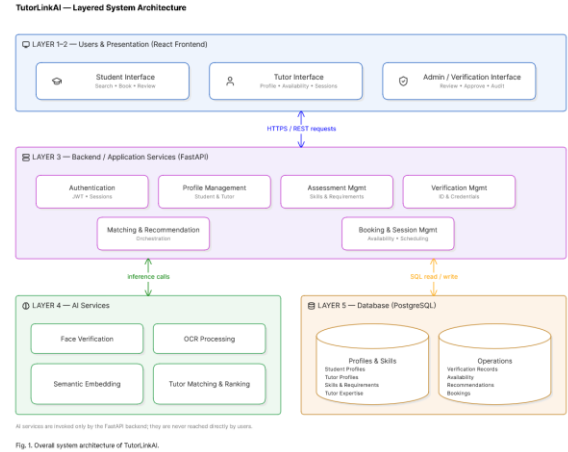


Fig. 1. Overall system architecture of TutorLinkAI.

B. Student and Tutor Profiling

The system provides separate processes for registration and profile management for students and tutors. The student profile captures information related to their academic requirements, subject skills, learning objectives, location, availability, and preferred budget. Tutor profiles contain information including subjects taught, areas of expertise, qualifications, experience, location, availability, pricing and verification status.

The collected information then forms the basis for subsequent assessment, verification, matching and recommendation processes. This information constructs the overall profile of the user.

1) Student Requirement Profile

The student's learning requirements are collected during the initial stage in the form of student registration and initial assessment. Rather than relying on selected subjects the system captures specific topics, areas of difficulty, skill level and learning objectives. This enables the matching component to identify the most suitable tutor according to the student's preferences and requirements.

2) Tutor Expertise Profile

Tutor expertise is represented using information such as subjects, topics, qualifications and teaching expertise. Both structured attributes and descriptive expertise information can be utilized during the matching process.

C. Tutor Verification

Tutor verification is incorporated to improve the reliability and credibility of tutor profiles. The verification module combines facial verification and document-based verification.

1) Face Verification

The system processes the tutor's submitted facial image to obtain a facial representation for identity comparison. OpenCV can be used for image processing, with InsightFace which is used for feature detection and facial structure detection. The resulting similarity information is used as one of the signals in determining the tutor's verification status [5], [12].

2) Certificate Verification

Uploaded qualification certificates are processed using Optical Character Recognition (OCR). PaddleOCR is used to extract relevant information from certificates such as the tutor's name, institution, qualification and other available textual fields. OCR-based extraction provides structured information that is compared against the tutor's profile information with stronger institutional validation as a future extension [4], [11].

D. AI Based Tutor Matching

The core component of TutorLinkAI is the tutor matching and recommendation system. Instead of relying solely on keyword-based or predefined filtering, the proposed approach considers both semantic similarity and structured compatibility between the student and tutor.

Student requirements and tutor expertise may be represented using sentence embeddings generated through Sentence Transformers[10], [13]. Semantic similarity between these representations can then be used to identify tutors whose expertise is relevant to the student's requirements.

E. Tutor Recommendation and Booking

Following the matching stage, suitable tutors are ordered according to how closely they satisfy the student's needs. Through the recommendation portal, students can review detailed tutor profiles covering qualifications, subject expertise, identity verification, location, pricing, and schedule to help them choose the most appropriate match.

The booking component maintains information regarding selected tutors, students, session schedules, and booking status. This provides an end-to-end workflow from tutor discovery to session scheduling.

VI. RESULTS AND DISCUSSIONS

The implementation of TutorLinkAI shows that it is possible to combine tutor onboarding and credential verification into a tutoring platform. The system that was built offers a step-by-step route for tutors to sign up, submit their qualifications and get checked before they appear as possible tutoring resources.

The verification module is where the most beneficial occurs. The system pulls out the text and examines them in, yet compared to what the tutor documented when they signed up. Very soon after we started developing it, we realized one thing: you can't have registration and verification happen at the same time. The tutor account does not prove the degree behind it is real. Splitting these into separate stages gives the platform a real basis for calling someone "verified" rather than just "signed up."

It's a major shift for parents in terms of decision making. They are not picking a tutor because a neighbor mentioned a name or someone lives two streets down the road. They can balance qualifications and verification status with location, which is a very different, and arguably more responsible, way to choose who teaches your kid.

But it is not without its flaws, OCR is only as good as the document that goes into it, and a blurry photo, bad handwriting or unusual certificate format can throw off the accuracy of extraction. The similarity check has a similar ceiling, it can tell you the extracted text doesn't match the registration form, but it has no way of confirming that the issuing institution itself is legitimate unless there's an external database to check against. So for now, the verification layer is an automated first pass at credibility not a replacement for an actual institutional check.

VII. CONCLUSION

TutorLinkAI solves a problem that has been poorly solved for a long time: how do you know if the tutor you are about to hire really knows what they say they know? The platform doesn't leave that question to word of mouth, it combines formal onboarding with automated checks on credentials and provides tutors with a real profile to stand behind.

Parents get something they never had before: a profile of the tutor, with actual verification status attached, not just a name and phone number passed along by a relative. This is the shift TutorLinkAI aims to achieve. Taking the guesswork out of choosing a tutor and turning it into an assessment a parent can make.

For tutors, the platform offers a real advantage. A tutor who has qualifications now has a clear way to prove those qualifications, not just a resume that nobody checks. Over time this will help separate qualified educators from people who are only good at self-promotion.

What this version shows is that it is possible to add credential verification to a tutoring platform that it can work without a person checking every paper by hand and that it gives

parents something real to see instead of a suggestion, from someone living three houses away. That is the gap TutorLinkAI aimed to fix and in that way it does what it was supposed to do.

VIII. FUTURE SCOPE

The current implementation of TutorLinkAI provides a foundation for automated tutor verification and recommendation. Further work can build upon this framework by improving credential verification, student performance analysis, security and personalized learning support.

1. Institutional Credential Verification: The system can be integrated with university, educational board and certification databases to verify tutor credentials directly from authorized sources. This mechanism can improve the reliability of the verification process and reduce submissions of invalid credentials and information[4], [11].
2. Advanced Document Fraud Detection: Image-forensics and machine-learning techniques can be incorporated to identify forged, manipulated or digitally altered documents by providing an additional layer of security in the tutor verification process[3].
3. Early Student Performance Prediction: Predictive machine learning models can be incorporated to analyze student learning patterns, assessment results and interaction data to identify students who may be at risk for poor performance. These early detection mechanisms can be helpful for tutors to provide the student with timely feedback and targeted support[1], [9].
4. Feedback and Rating Analysis: Natural Language Processing (NLP) techniques

can be used to analyse the student and parent feedback to tutoring sessions and identify common patterns in tutor performance. These extracted insights support students in making more informed decisions while choosing their tutors.

5. Continuous Tutor Verification: The platform can periodically reassess tutor credentials and profile information to ensure that verification remains current.
6. Scalable Cloud Deployment: Cloud infrastructure and containerization can be adopted to support an increasing number of students, tutors and concurrent verification requests. This improves the scalability and availability of the system.
7. Enhanced Security and Privacy: Future implementations can incorporate stronger encryption, role-based access control, audit logging, and privacy preserving storage mechanisms to protect sensitive credentials and academic documents[5], [12].
8. Multimodal AI-Based Doubt Solving: The platform can be extended with a multimodal AI module which is capable of processing text, images and other resources for providing the students with learning aid and resolving academic doubts. This system can provide interactive and context-aware learning support alongside tutor recommendations.

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